

Local Market Lab

Your private market analytics. Local. Secure.
Reproducible.

Version 0.8.0 — The open-source alternative to Bloomberg Terminal

Executive Summary

Local Market Lab (LML) is a locally-running, privacy-focused workbench for portfolio analytics, backtesting, scenario simulation, and AI-powered market prediction — all on your own machine, no cloud, no data sharing.

The Three Promises

Promise	Implementation
Privacy First	No telemetry, no external services, local SQLite
Institutional Methodology	CAGR, Sharpe, VaR, CVaR, Monte Carlo — professionally documented
Reproducibility	Every run has seed, timestamp, data hash

The Problem

Modern market analysis has three weaknesses:

Privacy: Most tools send your portfolio data to cloud servers. Your investment strategy becomes the product.

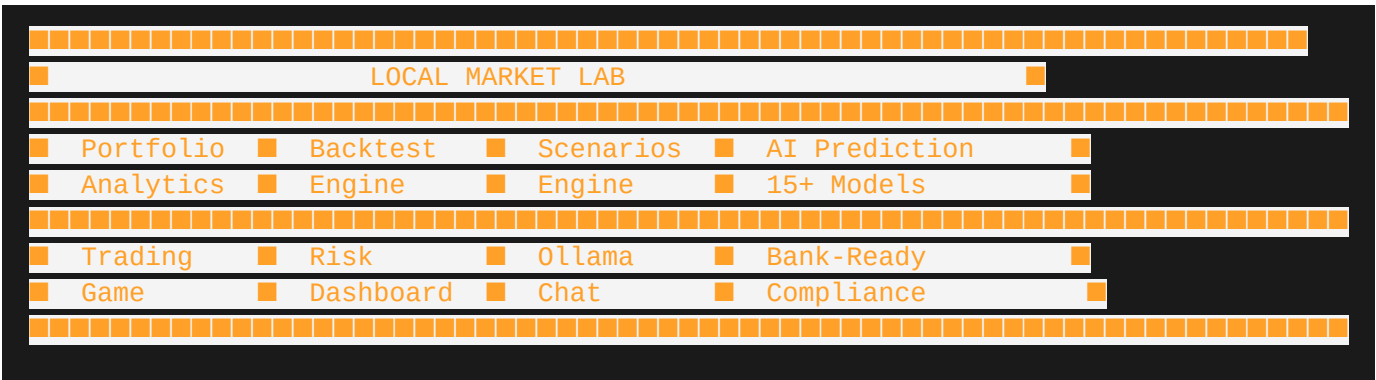
Black-Box Models: AI-powered predictions don't explain how they work. No control, no transparency.

Complexity: Professional tools (Bloomberg, Refinitiv) cost thousands of euros per month and require months of training.

LML solves all three.

The Solution

Local Market Lab combines six tools into one application:



Portfolio Analytics

- Real-time valuation with Decimal accuracy (no float errors)
- FX Policy with incomplete state (never silent 1:1 conversion)
- Corporate Actions: Splits, dividends, chronologically correct
- Benchmark comparison with Beta, Alpha, Tracking Error

Backtest Engine

- Event loop over aligned price series
- Strategies: Buy & Hold, Momentum, Mean Reversion, Periodic Rebalance
- Configurable fees and slippage
- Reproducibility manifests with seed and data hash

Scenarios

- Monte Carlo, Block-Bootstrap, Historical-Replay
- Percentiles (P05–P95), loss probability
- Explicit: **No forecast, only scenarios**

AI Prediction (15+ Models)

- **Basic Models:** Linear, Holt's ExpSmooth, ARIMA-like, Ensemble
- **Advanced:** Regime-Switching, Bayesian, Online Ensemble, Cross-Asset
- **Deep Learning:** LSTM, GRU with Backpropagation Through Time
- **Reinforcement Learning:** Q-Learning, DQN, REINFORCE
- **Genetic Optimization:** Feature Selection, Differential Evolution, NSGA-II

Trading Game

- Paper trading with virtual capital
- 6 Challenges (Beat Market, Low Volatility, Income Generator, ...)
- Leaderboard, equity curves, automatic mode

Risk Dashboard

- VaR, CVaR, Rolling Sharpe, Drawdown Series
- Performance Attribution, Correlation Matrix

- Correlation-Regime Detection

Technology

Component	Technology	Why
Backend	Python 3.11, FastAPI	Readability, ecosystem
Database	SQLite	Zero-config, portable
Web UI	HTML/Canvas (Bloomberg Style)	No dependencies
Desktop	PyQt6 + pyqtgraph	Native performance
Charts	Canvas 2D / pyqtgraph	Real-time, interactive
AI	100% numpy	No GPU, no CUDA
Tests	pytest	90+ tests, green
CI/CD	GitHub Actions, Docker	Automated

Use Cases

Retail Investors

- Understand portfolio development
- Quantify risks (VaR, Max Drawdown)

- Test strategies **before** real deployment

Traders

- Backtests with realistic costs
- Scenarios for different market phases
- Paper trading for strategy validation

Education

- Experience financial mathematics practically
- Understand AI/ML models (no black box)
- Learn privacy and reproducibility

Research

- Openly documented methodology
- All calculations reproducible
- Local data, no API dependency

Comparison

Feature	Local Market Lab	Bloomberg Terminal	Online Broker
Cost	Free (Open Source)	~€2,000/month	€0
Privacy	100% local	Cloud	Cloud
Methodology	Openly documented	Proprietary	Proprietary
AI Models	15+, local	External, black-box	None
Backtest	Fees, slippage, benchmark	Yes, but expensive	Limited
Reproducibility	Seed, hash, manifest	No	No
Offline capable	Yes	No	No
Windows App	Yes (PyQt6)	Yes (Native)	Web

Security & Compliance

Bank-Ready

- Audit Logger:** Append-only, SHA-256 checksums
- Data Integrity:** Automatic checksums for critical tables

- **GDPR Export:** JSON export of all user data
- **GDPR Deletion:** Anonymization with aggregate preservation
- **Compliance Report:** BaFin-compliant JSON

Security Principles

- No external API calls (except optional: Ollama)
- Rate Limiting (100 requests/minute/IP)
- Request ID for traceability
- Graceful shutdown with DB close

Roadmap

Version	Features
0.9	Walk-Forward Validation, Hyperparameter Tuning
1.0	Real market data (Yahoo Finance, Alpha Vantage)
1.1	Multiplayer Lobby, Team Challenges
1.2	Export: PDF, Excel, CSV Reports

2.0	Mobile App (iOS/Android)
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Team

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Call-to-Action

Start now — free, no registration, no cloud.

```
git clone https://github.com/mrmixx-max/local-market-lab.git
cd local-market-lab
pip install -e ".[dev]"
python -m apps.api
```

Or: Download the Windows installer from [GitHub Releases](#).

Private first. No cloud. No compromises.

LML v0.8.0 — The open-source alternative for privacy-focused market analytics.